Skills Summary

Finding and Fixing Data and Shape Mistakes

Use this reference while completing your worksheet or when studying.

Skill 1 — Checking a Scale

Rule: every step on a scale must go up by the *same* amount, and the gaps between the ticks must be the *same width*.

To find a missing label: (start value) + (how many steps across) × (step size).

Two things to check before reading any plot: are the labels evenly spaced in value, and are the ticks evenly spaced on the paper?

Example: A scale reads 2, 4, 6, 9. Which label is wrong, and what should it be?
Step 1. The steps between labels should all match, so I find the step size from the labels I trust and then rebuild the one that breaks the pattern.
Step 2. From 2 to 4 to 6 the step is 2, so the fourth label is $2 + 3 \times 2$, not 9.
the correct label is **8**

Try it: A scale reads 5, 10, 16. The step size is 5. What should the third label be?

check: **15**

Skill 2 — Counting the Right Thing

The most common data mistake: counting the numbers on the scale instead of the marks above them.

Numbers on the scale = the values. × marks = the things measured.

"How many were measured?" ⇒ add the column heights. "What is the difference between greatest and least?" ⇒ subtract two scale values.

Example: A plot has 3 marks above 1, 5 above 2 and 2 above 3. How many things were measured?
Step 1. Each thing measured put down exactly one mark, so the answer comes from the marks, not from the labels underneath them.
Step 2. Add the three column heights: $3 + 5 + 2$.
things measured = **10**

Try it: A plot has 4 marks above 1, 2 above 2 and 3 above 3. How many things were measured?

check: **9**

Skill 3 — Checking a Shape Against the Attribute

Rule: a shape belongs to a group only if it has *that group's attribute*. Having some other attribute does not get it in.

Four equal sides is not the same as four square corners. Four sides is not the same as two pairs of parallel sides.

So: read what the group asks for, then test that one thing on the shape.

Example: Someone says a rhombus is a square because both have four equal sides. How many square corners does the rhombus have?

Step 1. The claim is about being a square, and a square is defined by corners as well as sides, so I test the corners rather than recount the sides.

Step 2. Holding a card corner against each corner of the rhombus, not one of them fits: two are sharp and two are blunt.

square corners on a rhombus = **0**, so it is not a square

Try it: Someone puts a trapezoid in the "two pairs of parallel sides" group. How many pairs does a trapezoid really have?

check: **1**

(!) Watch out: finding the right answer is only half of a find-and-fix problem. The other half is saying *what* went wrong — "they counted labels instead of marks," "they added when the question said difference." Naming the mistake is what stops you making it.